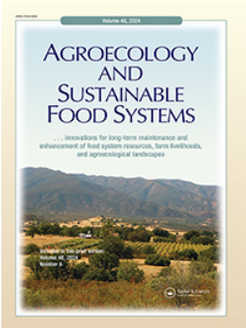

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Local food system resilience in the context of shocks and crises: vulnerabilities and responses of agroecology-based farmers in Peru, Germany, and the United States

Giovanna Chavez-Miguel[a,b,c], Janika Hämmerle[a], Antonio González[a], Chiara Canetti[a], Pia Gleich[a], Rebecca Lynn Halfast[a], Moritz Feuchter[a], Dominika Buszydlo[a], Laura Schwarz[a], Imke Scheepstra[a], Stef de Haan[c], Raul Ccanto[d], Stefan Sieber[a,b], and Michelle Bonatti[a,b]

[a]Resource Economics Group, Humboldt-Universität zu Berlin, Berlin, Germany; [b]Land Use and Governance, Leibniz Centre for Agricultural Landscape Research (ZALF E. V), Müncheberg, Germany; [c]Andean Initiative, International Potato Center (CIP), Lima, Peru; [d]Research and Innovation, Grupo Yanapai, Concepción, Perú

ABSTRACT

Multiple crises have disrupted food systems around the world. The shocks induced by the COVID-19 pandemic, coupled with increased energy prices and shortages of fertilizers and agricultural inputs, have affected farmers everywhere. By applying inductive, qualitative, and comparative research approaches, we investigate the impact of shocks and crises on diverse food systems. Using mixed qualitative methods, we integrate the voices of ($n = 25$) farmers belonging to agroecology-based local food systems in Peru, Germany, and the United States with the aims of: (i) understanding the different crisis scenarios; (ii) determining the vulnerabilities induced in farmers; and (iii) identifying the responses and coping mechanisms that farmers deploy in response to shocks. The results illustrate how farmers are affected by the combined effects of labor shortages, rising input costs, high energy prices, and climate change. Farmers' responses to stressors are then analyzed and compared using attributes of food system resilience. Based on our results, we argue that resilience-building is determined not only by the environment in which agroecology is embedded, but also by the responsive capacities of farmers. We emphasize the need for more proactive policies aimed at promoting agroecological and localized production to enhance the resilience of food systems against recurrent shocks and crises.

KEYWORDS
Social-ecological systems; responsive capacities

CONTACT Giovanna Chavez-Miguel Giovanna.Chavez-Miguel@zalf.de Resource Economics Group, Humboldt-Universität zu Berlin, Unter den Linden 6, Berlin 10117, Germany

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Introduction

Food systems in context of shocks and crises

Global food systems are suffering an increased number of logistical and economic disruptions due to interlinked shocks and crises (Vandermeer et al. 2018). In 2020, the shock caused by the coronavirus pandemic (Covid-19) disrupted food systems worldwide by restricting mobility for an extended period (Béné 2020; Bochtis et al. 2020). The restrictions caused delays in operations, disrupting regional and international food supply chains, leading to food shortages, food losses, food safety scares, and food price inflation (Bochtis et al. 2020). The immediate impacts of the COVID-19 shock are exacerbated by gradual changes or crises related to climate change, natural resource depletion, biodiversity loss, and political instability (Fan et al. 2021). In addition, global geopolitical conflicts, such as the war between Russia and Ukraine that broke out in early 2022, led not only to shortages of fertilizers and agricultural inputs, but also to increased energy and fuel prices, resulting in rapid increases in production costs and dramatic spikes in food prices (IPES-Food 2022).

The impact of the crises varied widely across regions (IPES-Food 2022): In the Global South, mobility restrictions had a major impact on socio-economic conditions. Market disruptions across regional, national, and international value chains quickly led to hunger and income losses among vulnerable rural and urban groups (Löhr et al. 2022). In Latin America, hardships often led to aggravated political crises (Eufemia et al. 2022), coupled with prevailing problems related to weak labor markets, infrastructure, and democratization processes. In contrast, food economies in the Global North did not experience the same type of disruptions, as maintaining logistical operations was a political priority, with essential food production and distribution activities

continuing despite government-imposed restrictions on mobility (Meuwissen et al. 2021). Actors within the food system were also disproportionately affected by the shocks and crises. Without minimizing the impact of COVID-19 on consumers, who suffered the economic consequences of the pandemic that affected their ability to access sufficient and nutritious food, agricultural producers were most vulnerable to the multiple effects of the crises. Located at the bottom of value chains and forced to maintain production and marketing activities regardless of the circumstances, farmers needed to endured hardship and uncertainty while developing coping strategies. Key challenges facing farmers related to market disruptions, labor shortages, rising input costs, high energy prices, and climate change. These problems lead to a vicious cycle of low production, low incomes, increased food insecurity (Löhr et al., 20–22), and, in some cases, to the abandonment of agriculture.

Strengthening food system resilience with agroecology

Given the impact that shocks and crises have had on local food systems, as well as their impact on farmers globally, there is a growing awareness of the need to construct *resilient food systems* capable of withstanding crisis scenarios (Blay-Palmer et al. 2020; HLPE 2020). More than ever, it is critical to rethink and transform current models of food production and distribution from a globally interconnected, long-value chain system to localized, contextualized, ecologically sound, and socially just food systems (Altieri and Nicholls 2020; Gliessman 2022).

Here, the concept of resilience has stimulated interdisciplinary research on sustainability worldwide (Holt-Giménez, Shattuck, and Van Lammeren 2021), leading to the development of various programs and initiatives aimed at strengthening the resilience of food systems in the face of uncertainty and change (West et al. 2014). Resilience is defined as the ability of a system to absorb perturbations, reorganize itself during change, and mitigate adversity by enabling the continuation of the systems over time (Walker et al. 2004). It can also be regarded as a measure of the *persistence of systems* in the face of complex and accumulating economic, social, environmental, and institutional shocks and stresses (Holling 1973). In the context of agri-food systems, the concept of *food system resilience* is emerging as a strategic tool to initiate food system transformation by addressing challenges related to sustainability and food security (Puma 2019; Zurek et al. 2022).

Agroecology is at the center of this needed transformation (Tittonell et al. 2021). As a scientific discipline, a set of practices, and a political movement, agroecology aims to improve the socio-ecological sustainability of farming systems (Holt-Giménez, Shattuck, and Van Lammeren 2021). Agroecology provides a holistic and human-centered framework for pursuing the design and implementation of the policies needed to transform the

food system (Gliessman 2022). However, there are different approaches within the agroecology movement reflecting the different conceptions of what sustainable and local food production should be (González De Molina and Lopez-Garcia 2021; Holt Giménez and Shattuck 2011). Accordingly, agroecological experiences around the world take many forms and are often fragmentary, small-scale, and driven by actors working mostly on a territorial or local scale (González De Molina and Lopez-Garcia 2021). This worldwide scattering of agroecology experiences represent niches of socio-ecological innovation helping advance local food system resilience (Anderson et al. 2019; Zurek et al. 2022).

Relevance and research objectives

To understand the broader agroecological landscape, it is of theoretical and empirical relevance to understand how resilience is built at the grassroots level by actors of agroecological food systems (López-García and González de Molina 2021). To this end, current empirical research on heterogeneous agroecological experiences emphasizes the importance of understanding the local specificities within which agroecology evolves. This is rightly so, since it is the spatial and socio-cultural particularities of each territory that help to elucidate the progress of agroecology in practice. However, experiences across highly diverse contexts are not yet consistently synthesized (Meyer 2020).

For this purpose, we examine the experiences of agroecology-based farmers in contexts of shocks and crises in Peru, Germany, and the United States (US). These countries were chosen due to the authors' ongoing fieldwork in 2022, which facilitated simultaneous data collection in these countries. Using inductive, qualitative, and comparative research approaches, as well as mixed ethnographic field methods, we (i) explore the socio-economic and political environment of the crisis scenarios in which farmers operate, (ii) identify the context-specific challenges facing farmers in each study area, and (iii) compare the responses and coping mechanisms that farmers deploy in face of adversity. To achieve these objectives, our research questions are formulated as follows:

(RQ1.1) How was the socio-economic and political environment in the context of recent shocks and crises in each of the study areas; and (RQ1.2) what vulnerabilities did this induce in agroecological farmers?

(RQ2) What responses and coping mechanisms did farmers deploy to maintain their agroecological production systems in these contexts? To what extent can these be understood as mechanisms for strengthening food system resilience?

To compare farmers' adaptive responses, we organize the results into resilience attributes identified in the literature (Table 2). Based on our findings, we discuss how food system resilience is built at the grassroots by food system actors, taking into account the contextual factors in each study area as well as the resilience capacities of farmers.

Methodology

Research design

A comparative, qualitative approach was chosen to explore the context-specific challenges faced by farmers practicing agroecology in contexts of shock in Peru, Germany, and the US. The comparison of these countries is justified based on the need to understand: i) how agroecology exists and evolves in different settings while concerning the contrasting contextual backgrounds of food systems in terms of institutional, socioeconomic, and cultural characteristics; ii) how actors behave and respond to shocks and crises in contrasting settings; and iii) how farmers overcome (or not) the different challenges that hinder their productive permanence, i.e., persistence, according to their capacities and the specificities of their socio-political and economic environments.

The research project emerged as part of an online seminar on *Research Methods in Agroecology*, taught by the lead author via Zoom, as part of the online Master's courses offered at Humboldt Universität zu Berlin in Germany. In this setting, the opportunity arose to compare these three case study countries. The authors had ongoing fieldwork activities, which allowed for simultaneous data collection during the uncertain and challenging times of 2022. Working individually and in teams, members of the research group conducted interviews at farms and farmers' markets in Peru, Germany, and the United States during June, July, and August 2022. Accordingly, a purposive sample was deliberately designed based on the authors' fieldwork duration and the routes of parallel research projects and internships.

Study areas

Interviews were conducted in six departments of the Andean region of Peru: Ancash, Apurímac, Cusco, Huancavelica, Huánuco, and Junín. In Germany, interviews were conducted in the federal states of Baden Württemberg, Bavaria, and Brandenburg, while in the United States (US), interviews were conducted during farm stays in California and Florida. The case study regions included in the sample are shown in the Figure 1 below.

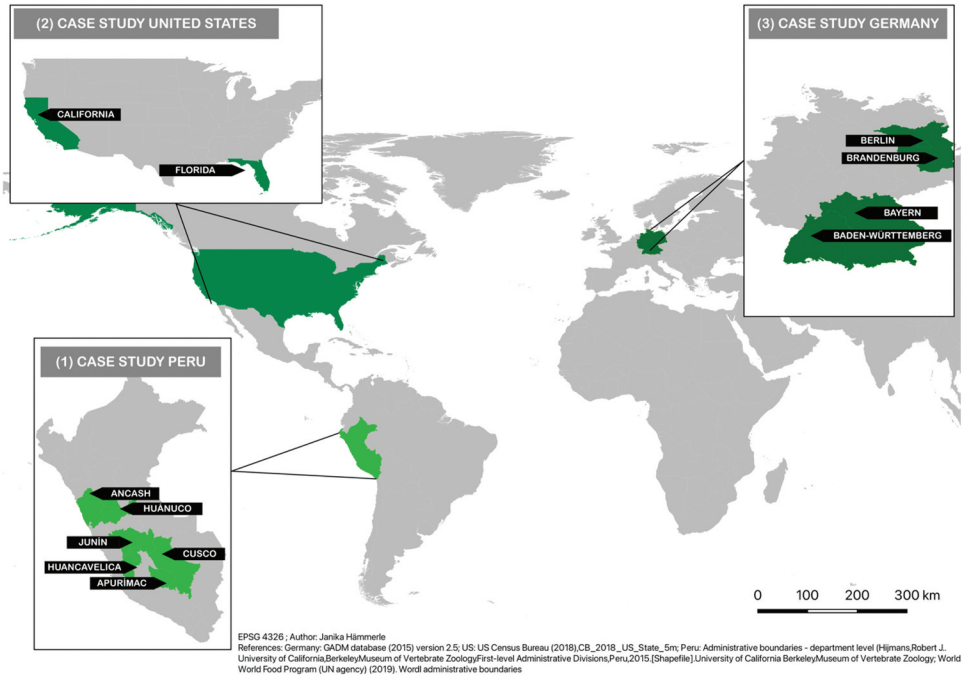

Figure 1. Map of selected case study areas. Source: Janika Hämmerle.

Sampling

The farms and interviewees in this study were selected based on their involvement in local food networks. Sampling methods varied between countries to identify relevant actors in heterogeneous food systems. In Peru, farmers participating in weekly local markets were interviewed using explorative sampling (see appendix 1). Fifteen semi-structured interviews were conducted at a total of eight markets with family farmers from the surrounding highlands and valleys, comprising primarily indigenous women ($n = 11$) who are the main marketers and male intermediaries ($n = 4$). Some farms were visited where we were able to verify that they maintain agroecological farming practices and preserve traditional local knowledge and agrobiodiversity. A description of the local agroecology of each region is seen in Table 1.

In Germany and the US, purposive sampling through online platforms helped identify farms that met our criteria. The four German farms are primarily sustainable permaculture projects linked to local market networks. We interviewed five respondents: 4 men and 1 woman from the four case study farms. In the US, we included four agroecological, regenerative or organic family farms with community-based and educational approaches, where we interviewed five respondents: 2 men and 3 women. Initially, more interviews were planned for Germany and the US, but due to time and resource constraints, the sample had to be reduced. Given that the Peruvian

Table 1. Description of the local agroecology by study area.

Case studies	Regions	Description of the local agroecology
Peru	Ancash, Apurímac, Cusco, Huancavelica, Huánuco, Junín	In Peru, the agroecology took the form of traditional subsistence agriculture practiced by family farmers living in remote areas of the highlands. At high altitudes (2,500–4,000 masl), farmers maintain agrobiodiversity by planting in terraces and mixing crops on steep terrains. In addition to tubers, farmers cultivate beans, horticulture, and root vegetables, while also keeping minor animal species, such as guinea pigs, and livestock, mainly alpacas, goats, and cattle. They preserve vast traditional knowledge related to the production and consumption of these goods (Mayer 2018). The mountainous topography and limited resources available to farmers in these areas often prevent the use of mechanization. Some farmers, even though they consider themselves to be agroecological, reported that they use reduced amounts of agrochemicals on their farms. This type of agroecology we characterize as *traditional agriculture*, in regards of the ancestral knowledge and practices that prevail around agrobiodiversity management.
Germany	Baden Württemberg, Bavaria, Brandenburg	In Germany, agroecology takes the form of peri-urban, community-supported agriculture projects. The farms are all located either inside cities or a few kilometers outside of urban areas. Farmers interviewed applied various alternative agriculture methods stemming from organic farming, permaculture, and regenerative agriculture. Their products, purposively chosen for selling, range from vegetables, fruits, and wild herbs to dairy and meat. We characterize these experiences as *market-oriented agroecology*, working toward the establishment of direct marketing channels to local grocery stores and farmer's markets.
US	California, Florida	In the US, the form of agroecology investigated included urban and peri-urban farms working toward implementing agroecological practices and strengthening local food systems. The farms surveyed ranged from nonprofit community-focused organic farms to for-profit farms. Farms were mainly small-sized, intensive market gardener productive systems of horticultural crops produced in greenhouses oriented to commercialization. Parallel to their productive activities, farms in the US also integrated activities including education and multi-product marketing as part of their business goals. We characterize these experiences as community-based, *market-oriented agroecology*.

Source: own compilation.

study was incorporated into a broader Ph.D. project, the sample from Peru was larger than the sample from the latter two countries. Despite differences in sample size, insights from all farmers remain relevant, warranting data analysis.able

Data collection

Farm visits, semi-structured in-depth interviews, and focus group discussions were employed in all study areas. A qualitative, semi-structured questionnaire was designed for our cross-case comparison study and adapted to each setting. It consisted of thematic blocks related to the crises studied, with respect to agricultural input prices, energy crisis, environmental change, and land use change. Open-ended questions were also used to identify other emerging

Table 2. Attributes of food system resilience.

Attributes	Meaning	Sources
Diversification	Productive diversification refers not only to the integration of multiple crops characteristic of agroecological farming, but also to the ability of food system actors to change the set of products they offer in the market. Moreover, diversification can also refer to the ability of actors to find different sources from which they obtain their inputs and supplies.	Tukamuhabwa et al. (2015); FAO (2021)
Entrepreneurship and innovation	Refers to the behavior of actors when they proactively adapt, take calculated risks, and innovate in response to internal and external stimuli. Entrepreneurship is considered to have the potential to improve actors' ability to anticipate and respond to stressors.	FAO (2021); Iza et al. (2019); Kangogo, Dentoni, and Bijman (2020)
Cooperation and Social Networks	Cooperation is an outcome of social capital; it refers to situations where food system actors seek win-win outcomes from working together. Cooperation within or between groups of actors is expected to reduce the negative impacts of shocks and crises.	Pingali, Alinovi, and Sutton (2005); Rose (2014); Downing et al. (2018); FAO (2021)
Farmer – buyer relationships/ Connectivity	Connectivity refers to the intensity and nature of relationships (vertical, horizontal, positive, negative) between different actors within and across socio-economic groups (farmers, traders, processors, and consumers). It is argued in the literature that connectivity would reduce the disturbances faced by producers and other actors along the food supply chain, thereby reducing their negative impact on operations and incomes.	Downing et al. (2018); Kangogo, Dentoni, and Bijman (2020);
Subjective resilience, knowledge, and experience	Refers to factors recognized as contributing to the construct of people's subjective resilience and influencing their choice of responses in the face of adversity, such as risk perception, self-efficacy, aspiration, or perseverance. This is a function of the knowledge and experience that farmers have.	Béné et al. (2019); FAO (2021)

Source: compiled by author.

themes. The interviews were conducted in Spanish, German, and English, respectively. We also applied complementary field methods from human geography and environmental anthropology, including participant observation and focused ethnographic approaches (Musante 2015). In Peru, we applied digital ethnographic methods. In Germany, data collection took place during extended farm stays, as well as through face-to-face and telephone interviews.

Data analysis

For the analysis of our data, we applied a multistep and inductive approach based on grounded theory (Glaser and Strauss 2017). Aiming to understand the contextual background within which agroecology develops and within which local food system actors operate, we applied a systematization methodology adapted from Chavez-Tafur, Hampson, and Thijssen (2006). The

results were first organized into two matrices by country (Appendices 4–9), one for farmers' vulnerabilities and a second for responses. Each of these matrices addresses four dimensions of sustainability (productive, economic, political, and socio-cultural). For the comparison, we structured our results in terms of (1) the impact of shocks and crises on farmers and (2) the responses of farmers (economic and productive). Appropriate conceptual and analytical frameworks were then constructed in an inductive manner (Glaser and Strauss 2017).

Analytical framework

For the analysis of the first part of our results (RQ1.1 and RQ1.2), we used a systematization methodology to cluster the impacts on food systems into turbulent and gradual changes (see Annex 3). As turbulent or abrupt changes, we considered the immediate impacts of shocks on food systems (Fan et al. 2021), i.e., disruption of market channels, labor shortages, high costs of agricultural inputs, and high energy prices. As gradual changes, we considered the effects of climate change, land use changes, and structural transformations in regional agricultural contexts, as well as the political environment of each study area. The vulnerabilities induced by these events for agroecology-based farmers are presented together in the first results section. The concept of vulnerability is commonly defined as a state that is the opposite of resilience (Adger 2000). A more refined definition is provided by Zurek et al. (2022), who describe food system vulnerability as the risk of exposure of (food) system actors to the impact of external influences, such as governance, policy, socio-technical, market, environmental, and economic drivers, as well as other shocks and stresses, which may lead to vicious cycles that jeopardize food security (p. 517). For the analysis of the second part of our findings (RQ 2), we evaluate our results in light of the food system resilience attributes or characteristics (Béné 2020) identified in the literature (Table 2).

Results

Impacts on food systems and induced farmer vulnerabilities

This first section outlines the main impacts that COVID-19 and other shocks and crises had on the food systems under study and the vulnerabilities induced in agroecological farmers, answering to RQ1.1 and RQ1.2 (see appendices 3–9 for more detail).

Market disruptions

In Peru, strict mobility restrictions were in place during the pandemic outbreak. One respondent explains: *"Only one person per family could go*

to the market. The police would not let you leave the house, not even to the farm. If you broke the rules, you went to jail for 24 hours and you were fined... or the police would beat you up" (P1). The restrictions had a significant negative impact on farmers' livelihoods in terms of lost income due to low sales, as many markets remained closed and farmers were unable to leave their communities. *"We saw our entire year's harvest rotting on our farms. We had also fattened our cattle, but we could not sell them"* (P8). As a result, there was a shortage of food (P6), leading to food price inflation, and *"prices have not come down since"* (P4). The price increases had a serious impact on food access and security. The impact of restrictions also affected food system actors in different ways. While intermediaries were given permits to move produce across regions, farmers were not (P1). As a result, *"the middlemen became abusive and paid an extremely low price when they bought our produce*" (P3). These conditions led to indebtedness and severe decapitalization of farmers (P1).

Meanwhile, the US and Germany did not experience major disruptions to food systems (G3, G4), as retailers and conventional distribution channels remained operational (G3). However, mobility restrictions affected US farms by preventing them from reaching storage facilities (US2). In contrast, farmers in Germany were unaffected by mobility restrictions: *"I was able to continue to drive everywhere, and the produce . . . continued to be picked up. Mobility in the surrounding area continued without problems*" (G4). Even though they did not experience a decrease in sales, except for one of the farmers whose produce is linked to restaurants, which remained closed (G3), farmers did experience delays in purchasing farm equipment (G3). In the US, farmers reported that only one niche market channel, namely farmers' markets, were restricted for a short period of time (US1–3). In both Germany and the US, farmers mentioned a positive outcome of the crises, namely an increased consumer awareness and demand for healthier and local food (G3, US1–3).

Labor shortage

In Peru, production was particularly affected by labor shortages, as farmers lacked the money to hire farm workers (P2). It is estimated that production has declined significantly and *"will most likely continue to decrease"* (P5), as farmers can no longer maintain production (P6). Owing to the very high production costs, including labor, "*only about half the hectares where potatoes used to be grown now produce*" (P5). As one trader explained, "*if farmers used to produce 100 sacks of potatoes, now only 50 sacks leave the communities*" (P6). The labor shortage did not affect the German farms because the farms were self-sufficient in their labor. In one case, more visitors were helping on the farm than before (G1). In the U.S., farms experienced labor shortages and interruptions in work. One had to lay off staff (US1), while another could not involve volunteers due to mobility restrictions (US2).

Rising costs of agricultural inputs

In Peru, production is heavily dependent on chemical fertilizers. However, since COVID-19 and the beginning of the war in Ukraine, Peru has experienced a rampant increase in the price of agricultural inputs: about 300% (P3, P8). The supply of these inputs has also suffered nationwide shortages. In past years, Peru used 450 million tons of fertilizer per year, but as of 2022, only 73 million tons were available (P6). As a result of high prices, many farmers have stopped producing for markets, are unable to plan for the next season, and/or are considering leaving agriculture (P2, P4, P7). In the US and Germany, high prices for conventional fertilizers did not affect our respondents because they use minimal or, in some cases, no chemical inputs (US1–4; G1–4). However, one farmer did notice an increase in the price of organic fertilizer and farm supplies due to panic buying: *"There was a period of impulse buying where all these people became home gardeners and farmers. Then our supplies became more expensive and harder to get"* (US2).

High energy prices

Global energy price increases affected farmers in all three areas. In Peru (P2, P3, P6, P7), many farmers were unable to get their produce to the markets due to increased transportation costs: "*Transportation costs are too high. . . we are forced to sell to middlemen who come to collect our potatoes directly from our farms at a much lower price. At some point, we preferred to keep our potatoes for our own consumption*" (P7). In addition, as fuel for tractors has become unaffordable, farmers are forced to revert to manual harvesting (P6). This has also led to planting delays (P3).

In Germany, farmers generally did not identify the energy crisis as a major problem for maintaining their operations (G1–4). Nevertheless, farmers have noticed an increase in the price of fuel for tractors as well as the cost of spare parts and repairs for agricultural machinery (G4). One respondent commented: *"Rising costs for energy and inputs, but stagnating prices for products"* (G1). Farmers also acknowledge that it may be too early to predict the future impact of the crises (G3). In the US, three farmers mentioned a noticeable cost increase in most farming supplies and delays in receiving the materials due to the increase in domestic fuel prices (US1, US2, US4). A few positive impacts were also mentioned: "*The energy crisis brought new incentives to build up photovoltaic capacities on the buildings; the price difference between organic-conventional products has decreased"(*G1).

Climate change

In the Peruvian Andes, rising temperatures have led to the melting of glaciers and a significant decrease in rainfall (P1, P7). This phenomenon has directly affected the productivity of family farms by reducing water availability, especially considering the lack of irrigation systems among

farmers (P1–5). To adapt to warmer temperatures, farmers have been forced to move to higher elevations to maintain potato production, as potatoes require cooler temperatures (P1, P2, P8). Rising temperatures have also led to an increase in pest infestations and extreme weather events affecting production (P4). In Germany, respondents observed changes in precipitation patterns since the 2010s, with summers becoming drier and winters milder (G1–4). These shifts have led to earlier emergence from winter dormancy for perennial crops, resulting in spring frost damage (G4). Prolonged droughts and water use restrictions inhibit crop and pasture growth (G1–4). Forestry land use has also been affected, with heat-induced growth inhibition and increased pest infestation leading to a significant loss of trees (G4). Despite the challenges, milder winters have facilitated the cultivation of new crop varieties (G2). Seasonal changes in planting activities will require adjustments, with consequential results: "*We might need to abandon vegetable cultivation on the parts of the farm that cannot be irrigated and instead grow high-quality, high-revenue vegetables on the small patch that can be irrigated*" (G3). In the US, Californian farms face recurrent fires, leading to reallocation of resources and labor (US2, US4). Unpredictable frosts also result in crop losses (US1, US4), while severe storms cause saltwater intrusions that severely impact crop growth or cause destruction (US1).

Farmer responses

This section describes the different economic and productive responses of farmers summarized in Table 3 (Appendixes 7–9), to the impacts outlined in the previous section, which are then clustered into *attributes of food system resilience* for ease of comparison.

Diversification

In Peru, those farmers who had the financial capacity to transform their production systems switched from growing potatoes to other crops, such as barley or fodder (P5), or integrated minor species into their farms: "*We are now breeding more animals than before, especially minor species, but also cattle, because we cannot live on our potatoes*" (P7). Some others decided to become full-time livestock farmers (P6). Effective productive diversification strategies were mainly implemented by farmers from Germany and the US (G2–4, US1). Farmers in Germany focused on increasing the resilience of their farms to climate stressors by diversifying the varieties grown, growing more resilient (often drought-resistant) varieties, diversifying production systems as much as possible so that overall production does not decline, to avoid crop failure, while maximizing the use of water and energy (G2, G3, G4).

Table 3. Results (RQ2): Farmers' responses to shocks and crises.

Case study	Economic responses	Productive responses
Peru	Diversification of income sources Creation of new selling points Innovations in logistics Activation of social networks Retaking practices of reciprocity Increased child labor Self-managed micro-loans to face decapitalization Substitution of potatoes for other crops	Adjusting farming practices Reviving agroecological practices Returning to manual labor Adopting animal breeding Restarting practices of reciprocity Shared labor *(faenas)* Reconfiguring productive systems Collective action
Germany	Opening new marketing channels Online marketing Selling on farm Strengthening social networks	Adjusting farming practices Maintaining organic practices Crop diversification Using drought resistant crops Abandoning vegetable cultivation Reducing farmland used Monitoring harvests yields and failures
US	Opening new marketing channels Increasing online marketing Activating communal/ family networks Strengthening self-sufficiency Diversifying product line Incorporating an educational approach	Productive planning Monitoring environmental conditions in cooperation with neighbors Using of more resistant varieties Using salt-tolerant plant species Increasing regenerative practices Housing external staff

Source: own compilation, based on appendices 7–9.

Entrepreneurship and innovation

In the United States, market restrictions encouraged farmers to adopt alternative distribution methods, primarily through online marketing, which allowed them to bypass commuting restrictions and fines (US1–4). Taking advantage of increased demand for local, fresh, and healthy products, they expanded their product range to include self-care products, ranging from beauty products and soaps to oils, flowers, and traditional remedies (US3, US4). German farmers also expanded their online presence, introducing not just on-farm pick-up options but also diversified product lines to maintain stable income streams (G1–4). In Peru, some farmers, particularly those involved in farmer associations, agroecological networks, agrobiodiversity conservation groups, and local markets, implemented logistical innovations during the economic downturn, including the relocation of collection points and initiatives in telecommunications and e-commerce (P4, P7, P8). Some agroecological potato producers also accessed niche markets by supplying a chip company at premium prices (P5).

Cooperation and social networks

Peruvian farmers relied heavily on community support during difficult times and reactivated social networks for support. The social fabric among families, within the communities and the marketplaces was strong: "*In these hard times, we can only help each other*" (P4). Farmers reactivated traditional reciprocity practices of exchanging labor and produce, known as *faenas* and *trueques*, i.e., barter, to overcome labor shortages (P5, P7, P8) and to acquire food without money (P3–8): "*Bartering is a beautiful custom that we keep, we don't lose it, and it's a good way to go on in difficult times*" (P3). Others report mutual support through self-managed micro-loans (P3, P4). Collective action was also visible as organized farmers and marketing groups created informal marketplaces in wastelands, occupied public spaces, and created itinerant marketplaces while formal markets remained closed (P4). In Germany, respondents benefited from personal and regional support networks, including associations, where helping each other and sharing equipment helped reduce operational costs (G3, G4). Similarly, in the US, community relationships enabled farmers to overcome challenges by working together to build storage facilities and to monitor groundwater levels and nearby fires (US2, US4).

Farmer – buyer relationships/connectivity

In Peru, weekly markets are key direct marketing channels that facilitate direct transactions between consumers and producers. Here, consumers and producers benefit from a "*better deal*"(P6). In difficult times, they also ensure food access (P6). However, there is a lack of consumer awareness of the nutritional and cultural value of farmers' produce, resulting in limited willingness to pay (P1–6). In the US, farmer-consumer relationships continued during the lockdown. Farmers benefited from the growing interest in home gardening by offering educational support and additional sources of income through classes and farm products (US1, US3). In Germany, farms responded to labor shortages caused by border closures, with regular consumers volunteering their labor in exchange for access to locally grown food (G1, G3).

Subjective resilience, knowledge, and experience

In Peru, family farmers who had increased the use of agrochemicals in recent decades are returning to agroecological farming due to the difficulty of accessing agrochemicals (P2, P7, P8). This transition, which is driven by necessity rather than choice (P7), involves the revival of traditional practices, for example, replacing chemical inputs with manure (P2). Highland farmers in Peru feel confident in their ability to undertake this agroecological transition because they maintain the necessary knowledge (P7, P8). In Germany, farmers have adapted to climate change and improved farm efficiency through observation and experimentation (G2, G3, G4): "*Understanding the causes of past crop failures is critical to increasing crop resilience*" (G2). They also adjust

production based on past season demand, improving predictability for future planting seasons (G1, G2). Understanding the causes of past crop failures is essential to increase crop resilience (G2). In the US, farmers are developing new knowledge related to agroecological practices, with some actively seeking to recover traditional knowledge (US2, US3). Some US farms feel they have extensive regenerative experience, making them less vulnerable to shocks (US2, US3).

Discussion

Agroecology and the construction of resilience

Based on our findings, this section explores the link between resilience and agroecology, highlighting the importance of agroecological actors in strengthening food system resilience from the grassroot level. As demonstrated by our results, actors in the agroecological food system are making considerable efforts to improve not just the resilience of their own farms, but also of the food system as a whole. Despite adversity, they persist in their practices to ensure long-term sustainability of food systems in the face of complex and accumulating economic, social, environmental, and institutional shocks and stresses (Holling 1973). It is important to note, however, that their efforts remain framed by the specific contexts in which they operate, and their outcomes are determined by their individual capacities and resources.

Enabling and constraining environments

As explained by Holt-Giménez, Shattuck, and Van Lammeren (2021), the ability of agroecology to build resilience extends beyond the biophysical to the socio-economic dimensions. This relates to the assertion that resilience-building is determined by the socio-ecological, economic, and political environment in which actors are immersed (Mathijs and Wauters 2020). Our findings show that, while all farmers faced significant challenges when maintaining agroecological practices in the face of adversity, the scenario in which farmers operated was very distinct, thereby influencing the resilience-building process.

The economic hardship faced by Peruvian farmers was nowhere close to the other two case studies. Peru was hard hit by the COVID-19 pandemic, reporting the highest mortality in the region (Chauvin 2021). The Peruvian government, in the attempt to prevent contagion, imposed severe mobility restrictions for several months. In enforcing these measures, the police and the military used physical force against civilians and fined those who failed to comply (Vargas et al. 2021). The imposed restrictions exacerbated existing socioeconomic inequalities among food system actors, as it led to abusive

behavior by middlemen toward family farmers (Malone, Cabana, and Zegarra 2021). With regards to the political and economic marginalization faced by farmers, the prevailing traditional form of agroecology appears to be under threat of becoming eroded. The erosion process has long been driven by low profitability due to the lack of formal market linkages for agrobiodiversity, low consumer purchasing power and awareness, alongside other macroeconomic factors that keep prices of farm products extremely low. These conditions were aggravated by crises and, in the absence of public support, led to the decapitalization of family farmers. In addition, overlapping socio-political problems related to state abandonment, social injustice, racism, and political instability keep farmers marginalized and without prospects for rural permanence. All in all, these conditions lead to the progressive abandonment of agriculture and contribute to the erosion of (traditional) agroecology.

In contrast, in the other two case studies – Germany and the United States – we found a more supportive environment for the construction of agroecology. In Germany and the US, the emergence of agroecology has paralleled the rise of the alternative agri-food movements, which grew out of concerns not only about on-farm sustainability, but also about food security, food safety, community, sustainability, and livelihoods in the food system (Fernandez et al. 2016). State interventions aimed at preventing the food system from collapsing were evident in the mobilization of state resources to support operational food chains, along with the provision of cash transfers to farmers (Davies et al. 2013). Thus, it can be argued that there was a more reliable social protection system in place for farmers (Béné et al. 2019; Davies et al. 2013). Although not all respondents used the cash transfers provided, either because of bureaucratic barriers or ineligibility, or because they did not need them, these social protection mechanisms were in place. Nonetheless, our results also show that, in Germany, agricultural policies and regulations do not necessarily alleviate the economic challenges faced by small-scale, organic farmers. Instead, they favor large-scale farming systems and emphasize energy-oriented land uses, such as photovoltaic systems and biofuels, over food production. Therefore, farmers also advocate greater support for those who produce sustainably and locally.

Actors' resilience capacities

Concurring with Béné (2020) and Constas et al. (2014), *resilience capacities*, also referred to as actors' resilience, determine the possible range of strategies and responses that actors may deploy for avoiding or reducing risk of long-term consequences in face of adversity (Bene 2020; Constas et al. 2014). These may be pursued at individual, household, or community levels, i.e., as forms of collective action (Darnhofer 2010), by slightly adjusting or fully changing their activities for either grasping an opportunity or mitigating the impact of

a threat (Zurek et al. 2022). For these to be successful, actors must be able to mobilize their assets, maintain flexibility, and instigate social organization, learning, and socio-cognitive capacities to actively shape their future (Cinner and Barnes 2019). Upon scrutinizing our results through these conceptual lenses, the resilience capacities of agroecological farmers exhibited considerable variation across the study regions. Notably, Peruvian farmers faced constraints due to limited financial and infrastructural resources, contrasting with their counterparts in Germany and the US. In this sense, the resilience capacities of Andean farmers were often hindered by financial and infrastructural limitations or did not always result in positive outcomes. Meanwhile, farmers in Germany and the US displayed greater resilience capacity, mainly because they were able to maintain their income sources through business innovations and absorb the financial impacts of the shocks with their assets.

To deepen our analysis, we draw on the concepts of appropriation and substitution proposed by Goodman, Sorj, and Wilkinson (1987) to examine agricultural transitions. According to Holt-Giménez, Shattuck, and Van Lammeren (2021), on the upstream (appropriation) side of agriculture, agroecology has the potential to build resilience by substituting conventional inputs, thereby reducing farmers' costs and vulnerability to price fluctuations in input markets. Our findings underscore how some agroecological farmers effectively reduced their vulnerability to input price increases by substituting unaffordable inputs with alternatives such as green manure, animal-based fertilizer, and biological pest control. In addition, farmers were better able to cope with rising costs if they diversified or adapted their product offerings in response to changing market demands. Notably, farmers with a clear understanding of their own market dynamics, showed greater stability in the face of these challenges. On the downstream (substitution) side, agroecology's potential for strengthening resilience lies in the existence of direct producer-consumer relations that are able to replace a complex of buyers and wholesalers (Holt-Giménez, Shattuck, and Van Lammeren 2021). Our results demonstrate how agroecological farmers in the Andes, who had organized for creating informal and itinerant market spaces in peripheral wastelands, while formal markets remained closed, were able to remain in market operation, thus countering the exploitative relationship that emerged between farmers and intermediaries. Moreover, those farmers who strengthened the links to their consumers and fostered farmer-buyer relationships, such as farmers in the US who chose to open their farms to consumers willing to visit or even volunteer on the farm, displayed a greater sense of resilience to the crises. This aspect was further illustrated by actors who showed a greater sense of entrepreneurship, i.e. those who proactively adapted to the rapidly changing market demands, took calculated risks, and innovated, such as farmers in Germany, who increased their online marketing, as well farmer organizations in the Andes that implemented direct-marketing schemes in cities. In light of our findings, we can say

that the appropriation and substitution responses of agroecological farmers do indeed contribute to the development of risk-averse farming practices and, hence, to resilience-building.

However, while we maintain that agroecology is effective in promoting resilience at the grassroots level, we acknowledge that its capacity falls short when it comes to broader market dynamics, territorial considerations, and government policies. Thus, we agree with Holt-Giménez, Shattuck, and Van Lammeren (2021) when they argue that agroecology alone cannot guarantee the resilience of farmers, communities or of whole agricultural systems. Therefore, it is crucial to ensure, at higher levels of governance, that the socio-economic and political environment is supportive for building resilient agroecological systems. Here, the state has a key role to play in enabling agroecology to flourish.

Methodological recommendations for further research

It is necessary to compare the mosaic of agroecological experiences emerging in different contexts to better understand the contributions of agroecology to the transformation of food systems. The comparison of these very different case studies, while not the most self-evident, provides valuable insights into how grassroots farmers working to ensure their own productive permanence contribute to the construction of more resilient food systems. Ethnographic and qualitative research methodologies are particularly valuable for capturing perceptions and sensitivities of farmers undergoing challenging processes of agroecological transitions. Particularly useful for these studies are systematization methodologies, which allow for exploratory data collection, and inductive analytical methods enabling the adaptation of conceptual frameworks. Future research in this area could benefit from larger samples and broader study regions.

Policy recommendations for strengthening local food system resilience

We emphasize the need for more proactive policies and measures aimed at strengthening the resilience of local food systems, as well as those that counteract the increasing tendency to rely on long food value chains and imports. Strengthening the resilience of food systems to future shocks and crises can be achieved by promoting agroecological production and localizing food supply. To this end, it is necessary to provide public support for the scaling of agroecological production, local food networks, and to improve the efficiency of existing short-distance distribution channels. It is also necessary to revalorize and revitalize traditional food systems, including local market networks and agrobiodiversity systems, as they contribute to the food security of rural communities. Finally, it is

imperative to recognize the scattered efforts of actors working to scale agroecology and alternative food networks, as well as to create coherence among them.

Conclusion

This article presents an empirical analysis of the heterogeneous experiences of farmers practicing agroecology and contributing to the construction of resilient food systems amidst recent shocks and crises. In 2020, the COVID-19 pandemic caused major food system shocks worldwide. The most significant shock was the near-universal response by governments to impose mobility restrictions, which disrupted logistics and markets worldwide. Soon after, the Russia-Ukraine war had a further impact on food systems with rising prices for agricultural inputs and energy. This study examines the socio-economic and political environments of the crisis scenarios in different regions and identifies the context-specific challenges faced by agroecological farmers in three study areas: Peru, Germany, and the United States. Our findings expose the multifaceted effects of the crises on food systems and show the different implications these had for farmers. In Peru, the effects of the shocks led to a vicious cycle of decapitalization and low incomes, declining production, increased food insecurity, and abandonment of farming. In Germany and the US, although farmers were able to maintain their activities, we find that farmers are in a more difficult position than before, mainly due to labor shortages and rising input prices. The different responses and coping mechanisms that farmers employed in the face of the crises were then compared along food system resilience attributes. The results show that successful strategies were implemented where there were stronger community ties and solidarity-based interactions between producers and consumers. In light of our findings, we then explored the relationship between resilience and agroecology. While we affirm the efficacy of agroecology in fostering resilience at the grassroots level, we recognize its limitations in addressing broader policy, market, and territorial dynamics. Hence, we advocate for actions to be taken at higher levels of governance that support the construction of more resilient agroecological and localized food systems.

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Ethics declaration

This study was a non-interventional study. Participation in the study was voluntary. All interviewees were informed about the purpose of this study and data protection issues by the enumerators prior to giving their consent orally at the beginning of each interview. Data is analyzed and presented in anonymous and aggregated form.

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